

# Christian Coduri

Turin, Italy

coduric



coduri



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https://coduri.cc/



## Employment History

11/2023 – 05/2024



**External Collaborator**, PWNX s.r.l., Remote

- Designed and implemented ten cryptography laboratories in *Python*, employing a "Capture the Flag" approach to enhance learning experiences.
- Developed more than thirty personalized one-on-one modules focused on *Linux Security* and *Cryptography Theory* for individualized learning.

05/2019 – 06/2019



**IT Intern**, SECAM s.p.a., Sondrio (SO).

## Education

11/2025 – now



**Ph.D. in Computer Engineering**, Polytechnic University of Turin

- **Research Area:** *Security in Artificial Intelligence Systems*.

10/2023 – 10/2025



**M.Sc. Cybersecurity Engineering**, Polytechnic University of Turin

- **Specialist track:** Cryptography expert.
- **Thesis title:** *Evaluating Backdoor Attacks Over Centralized and Distributed Medical Image Processing*.
- **Graduation Mark:** 110/110 cum laude | **GPA:** 29.65/30

10/2020 – 10/2023



**B.Sc. Computer Systems and Networks Security**, University of Milan

- **Thesis title:** *Advanced Techniques of Visual Cryptography and Steganography for Sharing of Medical Images*.
- **Graduation Mark:** 110/110 cum laude | **GPA:** 29.38/30

09/2015 – 07/2020



**High School Diploma in Computer Science**, ITIS E. Mattei (Sondrio)

- **Graduation Mark:** 96/100

## Additional Courses

10/2024 – 01/2025



**Automotive Cybersecurity**, Challenge@PoliTO by Stellantis

Collaborated in a multidisciplinary team to explore challenges related to future Autonomous Vehicles and V2X technology. My role mainly focused on identifying potential cybersecurity vulnerabilities and exploring theoretical solutions.

02/2023 – 05/2023



**National Cybersecurity Training Program**, Cybersecurity National Lab

Selected through the *CyberChallenge* initiative as one of 20 students from the University of Milan to join Italy's main government program for identifying and training future cybersecurity professionals.

Nov 2022



**International Cyber Conflicts**, The State University of New York (*Remote*)

Gained an understanding of international cybersecurity policies, as well as the psychological and sociopolitical factors influencing attackers and cyberattacks.

## Projects

- 2024 – 2025  **VisualCrypto**. A Python-based toolkit for *Visual Secret Sharing*, implementing cryptographic techniques that split a secret image into multiple shares. The project allows adding new cryptographic schemes and includes a web interface for ease of use. <https://github.com/coduri/VisualCrypto>.
- 2020 – 2022  **Get The Cookie**. A YouTube Channel with the goal of helping university and high school students understand computer science concepts (*in Italian*). Achieved over 1.2 million views, demonstrating the significant impact of the project. <https://www.youtube.com/GetTheCookie>.
- 2020 – 2021  **FitFocus**. An online fitness platform for nutrition tracking, progress monitoring, and workout planning. Included trainer-side dashboards for client management. Initially built on a *LAMP stack*, later migrated to *NodeJS* for improved performance.

## Publications

- 1 Christian Coduri and Stelvio Cimato. “A Framework for Secure Sharing of Medical Images Based on Visual Cryptography”. In: *IEEE 49th Computers, Software, and Applications Conference (COMPSAC Workshops 2025)*.  DOI: 10.1109/COMPSAC65507.2025.00280.
- 2 Christian Coduri et al. “Formal Verification and Mitigation of Vulnerabilities in the IMDGuard Protocol”. In: *IEEE 24th International Conference on Pervasive Computing and Communications (PerCom Workshops 2026)*.  DOI: 10.1109/PerComWorkshops68308.2026.11585400.

## Awards

- 2025 – 2026  **Fondazione Felix Leemann Scholarship** - Lugano (Switzerland). [6000 CHF]  
Awarded for academic excellence as a Swiss student from the Canton of Graubünden.
- Feb 2026  **Le Cento Tesi** by De Cifris Association. [link]  
Bachelor's thesis selected among the Best 100 Italian Cryptography Theses (2017–2024).
- Mar 2024  **STart STem Scholarship** by the Unione Industriali Torino. [3000€]
  - Selected as one of the top 64 first-year master's students in STEM disciplines at Polytechnic University and University of Turin.
  - Delivered 40 hours of **math academic support** to lower secondary school students.

## Professional Activities and Talks

- 21 May 2026  **Milanoday (Milan, Italy)**: Breaking Pacemaker Authentication with Formal Methods.
- 17 May 2022  **ISIS Gino Luzzatto (Portogruaro, Italy)**: What is Cybersecurity?